



UNSW
SYDNEY

MATH2601 – Higher Linear Algebra

Topic 6 – Eigenvalues and eigenvectors

Lecture 6.3 – Geometric properties and singular value decomposition

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Orthogonal matrices in $\mathbb{R}^3$

We have seen that any orthogonal (that is, real unitary) matrix $A \in M_{n,n}(\mathbb{R})$ is similar to a matrix of the form $I_a \oplus (-I_b) \oplus R(\alpha_1) \oplus \cdots \oplus R(\alpha_k)$ for $a + b + 2k = n$ and $\alpha_j \in (0, \pi)$ for all j .

The matrix $R(\alpha) = \begin{pmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{pmatrix}$ is known as a **rotation matrix** and these matrices encode useful geometric information.

We focus now on the particular case where $n = 3$. Recall that the group of orthogonal 3×3 matrices is called $O(3)$, and it has the subgroup $SO(3)$ which contains all orthogonal matrices with determinant 1.

Due to the above decomposition, we know any $A \in O(3)$ must be similar to a matrix of the form $\begin{pmatrix} \pm 1 & 0 & 0 \\ 0 & \cos(\alpha) & -\sin(\alpha) \\ 0 & \sin(\alpha) & \cos(\alpha) \end{pmatrix}$ for some $\alpha \in [0, \pi]$.

Fact. If the first entry is 1, then $\det(A) = 1$ and the map $\mathbf{v} \mapsto A\mathbf{v}$ represents a **rotation** by the angle α around the axis E_1 .

Fact. If the first entry is -1 , then $\det(A) = -1$ and the map $\mathbf{v} \mapsto A\mathbf{v}$ represents a **reflection** through the plane with normal E_{-1} , composed with a **rotation** by the angle α around the axis E_{-1} .

Example 1

Exercise. Interpret geometrically the action of the matrix

$$A = \frac{1}{9} \begin{pmatrix} 4 & 7 & -4 \\ 1 & 4 & 8 \\ 8 & -4 & 1 \end{pmatrix} \text{ on } \mathbb{R}^3.$$

Solution. We can confirm that A is orthogonal by checking that as vectors, each column has norm 1, and the dot product of each pair of distinct column vectors is 0. Since $A \in O(3)$, it must represent a rotation or a reflection composed with a rotation. Since $\det(A) = 1$, we know a real eigenvalue of A is 1 and that A is just a rotation.

Since $\text{tr}(A) = 1$ and $\text{tr}(I_1 \oplus R(\alpha)) = 1 + 2\cos(\alpha)$, we know that $\cos(\alpha) = 0$ and so $\alpha = \frac{\pi}{2}$.

Since $E_1 = \text{span} \left\{ \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} \right\}$, we can conclude that the action of the matrix on $\mathbb{R}^3$ rotates any vector by $\frac{\pi}{2}$ around the axis E_1 .

Note we could also have confirmed that $A \in SO(3)$ by checking that the third column is equal to the **cross-product** of the first column with the second column.

Example 2

Exercise. Interpret geometrically the action of the matrix

$$A = \frac{1}{9} \begin{pmatrix} 4 & -7 & -4 \\ 1 & -4 & 8 \\ 8 & 4 & 1 \end{pmatrix} \text{ on } \mathbb{R}^3.$$

Solution. Since this matrix is the same as the previous example but with the middle column multiplied by -1 , it must still be orthogonal, and it must have determinant -1 . Since $\det(A) = -1$, we know a real eigenvalue of A is -1 and that A is a reflection composed with a rotation.

Since $\operatorname{tr}(A) = \frac{1}{9}$ and $\operatorname{tr}((-I_1) \oplus R(\alpha)) = -1 + 2\cos(\alpha)$, we know that $\cos(\alpha) = \frac{5}{9}$ and so $\alpha = \arccos(\frac{5}{9})$.

Since $E_{-1} = \operatorname{span} \left\{ \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix} \right\}$, we can conclude that the action of the matrix on $\mathbb{R}^3$ reflects any vector in the plane normal to E_{-1} , and then rotates it by $\arccos(\frac{5}{9})$ around the axis E_{-1} .

Note we could also have confirmed that $A \in O(3)$ and $\det(A) = -1$ by checking that the third column is the **negative** of the cross-product of the first column with the second column.

Conics

A **conic section** is any curve in $\mathbb{R}^2$ whose equation contains expressions of degree at most 2 in the variables x and y . This includes parabolas (e.g. $y = x^2$), ellipses (e.g. $x^2 + 2y^2 = 1$), and hyperbolas (e.g. $x^2 - y^2 = 1$).

If a conic equation is of the form $ax^2 + bxy + cy^2 = d$, it can be rewritten as $\mathbf{v}^T A \mathbf{v} = d$ where $\mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2$ and $A = \begin{pmatrix} a & \frac{b}{2} \\ \frac{b}{2} & c \end{pmatrix}$ is **symmetric**.

Fact. The signs of the **eigenvalues** of A determine the shape of the conic (ignoring degenerate cases). Specifically:

- If both eigenvalues are **positive**, then the curve is an **ellipse**.
- If the eigenvalues have **opposite sign**, then the curve is a **hyperbola**.
- If one of the eigenvalues is **zero**, then the curve is a **parabola**. **Note** that without any linear terms, the equation $ax^2 + bxy + cy^2 = d$ will always form a degenerate parabola, which is **two parallel lines** (for $d > 0$).

The reason for this is that for the eigenvalues λ and μ of A , because A is symmetric (self-adjoint), we know the corresponding eigenspaces E_λ and E_μ are orthogonal. So writing $D = P^{-1}AP$ as usual for diagonal D and unitary P , we can set new coordinates $\mathbf{w} = \begin{pmatrix} X & Y \end{pmatrix}^T = P^{-1}\mathbf{v}$.

Then $d = \mathbf{v}^T A \mathbf{v} = \mathbf{v}^T P D P^{-1} \mathbf{v} = (P^T \mathbf{v})^T D \mathbf{w} = \mathbf{w}^T D \mathbf{w} = \lambda X^2 + \mu Y^2$.

Conics

The fact that P is unitary guarantees that the change of coordinates preserves the shape of the curve, since we showed earlier that orthogonal/unitary matrices map orthonormal bases to orthonormal bases.

Any geometric property of the conic can be found in these “nicer” coordinates and then translated back via the transformation $\mathbf{v} = P\mathbf{w}$.

Exercise. Describe the curve $8x^2 + 12xy + 3y^2 = 48$.

Solution. We can write the equation as $\mathbf{v}^T A \mathbf{v}$ where $A = \begin{pmatrix} 8 & 6 \\ 6 & 3 \end{pmatrix}$. This matrix has eigenvalues 12 and -1 , so since they have opposite signs, the curve is a hyperbola (with the same shape as $12X^2 - Y^2 = 48$).

We can find $D = P^{-1}AP$ where $D = \begin{pmatrix} 12 & 0 \\ 0 & -1 \end{pmatrix}$ and $P = \frac{1}{\sqrt{13}} \begin{pmatrix} 3 & -2 \\ 2 & 3 \end{pmatrix}$.

Since the principal axes in the new coordinate system are the X - and Y -axes, the principal axes of the original hyperbola are the lines through $P \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $P \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, which up to a scalar factor are $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$.

(Similarly, the points closest to the origin are $P \begin{pmatrix} \pm 2 \\ 0 \end{pmatrix} = \pm \frac{1}{\sqrt{13}} \begin{pmatrix} 6 \\ 4 \end{pmatrix}$.)

Singular value decomposition

We have seen various ways to decompose matrices, including QR-factorisation and diagonalisation (spectral decomposition). Another computationally useful matrix decomposition is known as the singular value decomposition (SVD):

Definition. Given $A \in M_{m,n}(\mathbb{C})$, a **singular value decomposition** of A is the factorisation $A = U\Sigma V^*$ where $U \in M_m(\mathbb{C})$ is **unitary**, $V \in M_n(\mathbb{C})$ is **unitary**, and $\Sigma \in M_{m,n}(\mathbb{C})$ is **diagonal** in the sense that the only non-zero entries are at positions (i, i) for $1 \leq i \leq r = \text{rank}(A)$.

Furthermore, the diagonal entries of Σ , called the **singular values**, are all non-negative and can be arranged in decreasing order from left to right. This can be written as $\sigma_{1,1} \geq \cdots \geq \sigma_{r,r} \geq 0$ or just $\sigma_1 \geq \cdots \geq \sigma_r \geq 0$.

We call the columns of U **left singular vectors** and the columns of V **right singular vectors**.

Fact. Such a SVD factorisation exists for any matrix $A \in M_{m,n}(\mathbb{C})$, and the singular values are unique for each A (we will prove this later).

Constructing SVDs

Given any $A \in M_{m,n}(\mathbb{C})$, we can find its singular value decomposition by following these steps:

- Find A^*A .
- Find the eigenvalues of A^*A (with multiplicity) and list them in decreasing order $\lambda_1, \dots, \lambda_n$. Set the singular values of Σ to be $\sigma_i = \sqrt{\lambda_i}$ for each $1 \leq i \leq r = \text{rank}(A)$.
- Set V to be a unitary matrix P for which $P^{-1}(A^*A)P = P^*A^*AP = D$ where $D = \Sigma^*\Sigma$ is diagonal.
- Writing $\mathbf{u}_i$ and $\mathbf{v}_i$ for the i th columns of U and V respectively, construct U by setting $\mathbf{u}_i = \sigma_i^{-1}A\mathbf{v}_i$ for each $1 \leq i \leq r$, and fill out any remaining columns of U by extending to an orthonormal basis of $\mathbb{C}^m$.

Example. Find a singular value decomposition for $A = \begin{pmatrix} 1 & 2 \end{pmatrix}$.

Solution. We have $A^*A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$, so $D = P^{-1}A^*AP$ where $D = \begin{pmatrix} 5 & 0 \\ 0 & 0 \end{pmatrix}$

and $P = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}$ is unitary. So $\mathbf{u}_1 = (\sqrt{5})^{-1} \begin{pmatrix} 1 & 2 \end{pmatrix} \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = (1)$,

and $A = U\Sigma V^*$ where $U = (1)$, $\Sigma = (\sqrt{5} \ 0)$, and $V = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}$.

Example

Exercise. Find a singular value decomposition for $A = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \end{pmatrix}$.

Solution. We have $A^*A = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ -1 & 1 & 2 \end{pmatrix}$, with eigenvalues 3, 1, and 0.

Diagonalising, we have $D = P^{-1}A^*AP$ where $D = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ and

$P = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 & \sqrt{3} & \sqrt{2} \\ -1 & \sqrt{3} & -\sqrt{2} \\ -2 & 0 & \sqrt{2} \end{pmatrix}$ is unitary.

So $\mathbf{u}_1 = (\sqrt{3})^{-1} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \end{pmatrix} \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ and

$\mathbf{u}_2 = 1^{-1} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

So $A = U\Sigma V^*$ where $U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$, $\Sigma = \begin{pmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$, and $V = P$.

Existence of SVDs

Lemma. If $A \in M_{n,n}(\mathbb{F})$ is diagonalisable, then the rank of A is equal to the number of non-zero eigenvalues of A (counting algebraic multiplicities).

Proof. Since A is diagonalisable, each eigenvalue must have equal algebraic and geometric multiplicities. Note that if 0 is an eigenvalue of A , then $E_0 = \ker(A - 0I) = \ker(A)$, so $\text{nullity}(A)$ is the geometric multiplicity of 0. So by the rank-nullity theorem, the rank of A must be the sum of the geometric multiplicities of all the non-zero eigenvalues.

Lemma. Given $A \in M_{m,n}(\mathbb{C})$, the matrix A^*A is self-adjoint and its eigenvalues are all non-negative.

Proof. Writing $\mathbf{a}_i$ for the i th column of A , for all $i \neq j$ we have $(A^*A)_{i,j} = \mathbf{a}_i^* \mathbf{a}_j = (\mathbf{a}_j^* \mathbf{a}_i)^T = \overline{\mathbf{a}_j^* \mathbf{a}_i} = \overline{(A^*A)_{j,i}}$, so A^*A is self-adjoint. Furthermore, if λ is an eigenvalue for A^*A with corresponding eigenvector $\mathbf{v}$, we have $0 \leq \|A\mathbf{x}\|^2 = (A\mathbf{x})^*(A\mathbf{x}) = \mathbf{x}^* A^* A \mathbf{x} = \mathbf{x}^* \lambda \mathbf{x} = \lambda \|\mathbf{x}\|^2$, so $\lambda \geq 0$.

Lemma. Given $A \in M_{m,n}(\mathbb{C})$, we have $\ker(A^*A) = \ker(A)$ and $\text{rank}(A^*A) = \text{rank}(A)$.

Proof. Note that if $A\mathbf{v} = \mathbf{0}$ then $A^*A\mathbf{v} = \mathbf{0}$, so $\ker(A) \subseteq \ker(A^*A)$. Also, if $A^*A\mathbf{v} = \mathbf{0}$ then $\|A\mathbf{v}\|^2 = \mathbf{v}^* A^* A \mathbf{v} = \mathbf{v}^* \mathbf{0} = 0$, so $A\mathbf{v} = \mathbf{0}$ by positive definiteness and so $\ker(A^*A) \subseteq \ker(A)$. Thus $\ker(A^*A) = \ker(A)$, and by the rank-nullity theorem $\text{rank}(A^*A) = \text{rank}(A)$.

Existence of SVDs

Theorem. For any $A \in M_{m,n}(\mathbb{C})$, there exists a singular value decomposition, and the singular values are unique for each A .

Proof. Let $r = \text{rank}(A)$. By the previous lemmas, A^*A has exactly r non-negative eigenvalues $\lambda_1, \dots, \lambda_r$. Arranging these in decreasing order, we can set $\sigma_i = \sqrt{\lambda_i}$ for each i and make these the diagonal entries of $\Sigma \in M_{m,n}(\mathbb{C})$. Then set $V \in M_n(\mathbb{C})$ to be a unitary matrix P for which $P^{-1}(A^*A)P = P^*A^*AP = D$ where $D = \Sigma^*\Sigma$ is diagonal. Writing $\mathbf{u}_i$ and $\mathbf{v}_i$ for the i th columns of U and V respectively, construct $U \in M_m(\mathbb{C})$ by setting $\mathbf{u}_i = \sigma_i^{-1}A\mathbf{v}_i$ for each $1 \leq i \leq r$, and fill out any remaining columns of U by extending to an orthonormal basis of $\mathbb{C}^m$.

By construction, $U\Sigma = AV$, so $A = U\Sigma V^*$. Also, U is unitary since for each i, j we have $(U^*U)_{i,j} = \mathbf{u}_j^*\mathbf{u}_i = \sigma_j^{-1}\sigma_i^{-1}\mathbf{v}_j^*A^*A\mathbf{v}_i = \sigma_j^{-1}\sigma_i^{-1}(\sigma_j\mathbf{e}_j^T)(\sigma_i\mathbf{e}_i)$ (since $V^*A^*AV = \Sigma^*\Sigma$), which is 1 whenever $i = j$ and 0 otherwise, meaning $U^*U = I$.

To prove uniqueness, note that if $A = U\Sigma V^*$ as described then $A^*A = V\Sigma^*U^*U\Sigma V^* = V\Sigma^*\Sigma V^*$. Since $\Sigma^*\Sigma$ is diagonal and A is similar to it, its diagonal entries must be the eigenvalues of A , so after arranging in decreasing order, the singular values are fixed.

Properties of SVDs

Lemma. For any $A \in M_{m,n}(\mathbb{C})$ with rank r , the first r left singular vectors (columns of U in a SVD) form an orthonormal basis for $\text{im}(A)$, and the last $n - r$ right singular vectors (columns of V in a SVD) form an orthonormal basis for $\ker(A)$.

Proof. The last $n - r$ right singular vectors correspond with 0-eigenvalues of A , that is, elements of the kernel of A , and are orthonormal by construction. The first r left singular vectors are clearly orthonormal and elements of the image of A by construction, and must span $\text{im}(A)$ since their size is the rank of A .

Lemma. For any $A \in M_{m,n}(\mathbb{C})$ with singular values σ_i , we have $\sum \sigma_i^2 = \text{tr}(A^*A)$, which is $\|A\|^2$ under the standard matrix inner product $\langle A, B \rangle = \text{tr}(A^*B)$.

Proof. This follows immediately from the fact the trace of a matrix is the sum of its eigenvalues (counting algebraic multiplicity), and that the singular values are the square roots of the eigenvalues.

Reduced SVDs

Definition. Given $A \in M_{m,n}(\mathbb{C})$ has rank r , the **reduced singular value decomposition** for A is a decomposition $A = \hat{U}\hat{\Sigma}\hat{V}^*$ where $\hat{\Sigma} \in M_r(\mathbb{C})$ is **diagonal** and **invertible**, and $\hat{U} \in M_{m,r}(\mathbb{C})$ and $\hat{V} \in M_{n,r}$ have **orthonormal columns**.

The reduced SVD is constructed by first finding the standard SVD, then restricting Σ to size $r \times r$ and removing the last $m - r$ columns from U and the last $n - r$ columns from V .

Example. From the previous example with $A = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \end{pmatrix}$, the reduced SVD is given by $A = \hat{U}\hat{\Sigma}\hat{V}^*$ where $\hat{U} = U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$,

$$\hat{\Sigma} = \begin{pmatrix} \sqrt{3} & 0 \\ 0 & 1 \end{pmatrix}, \text{ and } \hat{V} = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 & \sqrt{3} \\ -1 & \sqrt{3} \\ -2 & 0 \end{pmatrix}.$$

Pseudoinverses

Definition. Given $A \in M_{m,n}(\mathbb{C})$ with a reduced singular value decomposition $\hat{U}\hat{\Sigma}\hat{V}^*$, the **pseudoinverse** of A , written A^+ , is given by $A^+ = \hat{V}\hat{\Sigma}^{-1}\hat{U}^*$.

Recall that for the system of equations $A\mathbf{x} = \mathbf{b}$, if the columns of A are linearly independent then the least squares solution is unique and given by $\mathbf{x} = (A^*A)^{-1}A^*\mathbf{b}$.

Theorem. If $A \in M_{m,n}(\mathbb{C})$ has rank n , the **least squares solution** to $A\mathbf{x} = \mathbf{b}$ is given by $\mathbf{x} = A^+\mathbf{b}$.

Proof. First note that $\Sigma^* = \Sigma$. Since A is full rank, A^*A is full rank and hence invertible, and V is square and hence invertible. So

$$A^*A = \hat{V}\hat{\Sigma}^*\hat{U}^*\hat{U}\hat{\Sigma}\hat{V}^* = \hat{V}\hat{\Sigma}\hat{\Sigma}\hat{V}^*,$$

and hence

$$\mathbf{x} = (A^*A)^{-1}A^*\mathbf{b} = (\hat{V}\hat{\Sigma}^{-1}\Sigma^{-1}\hat{V}^*)(\hat{V}\hat{\Sigma}\hat{U}^*)\mathbf{b} = \hat{V}\hat{\Sigma}^{-1}\hat{U}^*\mathbf{b} = A^+\mathbf{b}.$$